Milestone 4 Progress Evaluation

Project Ragnarok: Post-Quantum Cybersecurity with Lightning Data

Group Members

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Advisor / Client

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- Dr. Amitabh Nag

Institution: Florida Institute of Technology – Senior Design

1. **Progress of Current Milestone (Progress Matrix)**

Task	Completion	Aidan	Gianni	Joanna	To Do
1. Test Rust Server on Raspberry Pi	90%	10%	70%	10%	Integrate database importation and ensure expected behavior continues
2. Data fetchers	80%	80%	0%	0%	Finalize automated fetch scripts and error handling for dropped API connections.
3. Entropy analysis and maximization	50%	50%	0%	0%	Complete statistical randomness testing (NIST SP 800-22 suite) on normalized datasets.
4. Gather high entropy data sets with over 10^6 samples	100%	33%	33%	33%	Task completed. Datasets stored and formatted for model ingest.

2. **Discussion of Accomplished Tasks**

Task 1: Test Rust Server on Raspberry Pi

The Rust server has been successfully ported to the Raspberry Pi hardware environment. Cross-compilation target binaries were generated and verified. The server runs as expected in the local ARM architecture; however, automated data importers and initial database hydration scripts on server startup have yet to be executed on the Pi hardware.

Task 2: Data Fetchers

Automated data fetching pipelines were developed to continuously ingest lightning, solar wind, and seismic data from external sensors and public API streams. The fetchers normalize raw lightning strike events (including amplitude, location, microsecond timestamp, and stroke multi-peak parameters) into structured CSV data ready for post-processing.

Task 3: Entropy Analysis and Maximization

Initial statistical analysis was conducted on raw lightning telemetry streams to extract high-entropy seed strings for post-quantum cryptographic key generation. hash-based extractors like sha-3 and toplitz hashing are going to be used to satisfy the leftover hashing lemma and ensure high entropy random number we have calculated the min entropy need to produce 256 bit uniformly random number to be 384 bits with an error 2^-64 from uniform

Task 4: Gather High Entropy Data Sets ($> 10^6$ Samples)

The team successfully aggregated over 1,000,000 valid lightning telemetry samples across multi-region datasets. The data was cleaned, validated for missing entries, formatted into standardized CSV collections, and deposited into shared persistent storage for local server testing and entropy validation.

3. Team Member Contributions

Aidan Nelappana: Led the development of the automated data fetcher pipeline. Implemented data extraction logic from telemetry sources and spearheaded the entropy extraction, normalization, and statistical analysis methodologies to ensure cryptographically suitable seed generation.

Gianni Bubb: I took the responsibility of ensuring the Pi could run the server. It was a challenge figuring out how to port the server to the Pi, as I had never attempted cross-compilation before. After quite a bit of time, I managed to port the server to the Pi via transferring the file and the database CSV through a flash drive. Currently, the server has been running for a while, but I have not figured out how to handle the database import on server startup on the Pi.

Joanna Zhang: Coordinated dataset structure and database schema preparation. Contributed to raw data collection and validation for the $10^6 +$ sample repository, and assisted in benchmarking server response behaviors during initial startup tests.

4. Plan for Next Milestone (Task Matrix)

Task	Aidan	Gianni	Joanna
1. Fully implement and secure the Python API using token-based authentication (OAuth2/JWT)	20%	60%	20%
2. Integrate NIST-approved Post-Quantum Cryptography (PQC) schemes into the key generation pipeline	60%	20%	20%
3. Implement end-to-end database security and access control mechanisms for raw lightning telemetry data	20%	20%	60%

5. **Discussion of Planned Tasks**

Task 1: Secure Python API (OAuth2/JWT): Develop secure endpoints for querying entropy metrics and generating keys. Implement OAuth2/JWT authentication, rate limiting, and role-based access control (RBAC) to ensure unauthorized clients cannot trigger key regeneration or intercept stream access.

Task 2: NIST-Approved PQC Integration: Integrate post-quantum algorithms (such as ML-KEM / Kyber for key encapsulation and ML-DSA / Dilithium for digital signatures) into the primary key generation pipeline, seeding the implementations directly with maximized lightning entropy output.

Task 3: End-to-End Database Security & Access Control: Implement encryption at rest (AES-256-GCM) for stored physical datasets, secure DB connection channels on the Raspberry Pi using TLS/mTLS, and configure strict user roles to isolate database administrative actions from standard server reads.

Meetings

Client Meeting Date(s)
, 2026

Client Feedback

Faculty Advisor Meeting Date(s)

, 2026

Faculty Advisor Feedback

Task 1:

Task 2:

Task 3:

Task 4:

Faculty Advisor Signature: _____ Date: _____

Evaluation by Faculty Advisor

- Faculty Advisor: detach and return this page to Dr. Chan (HC 209) or email the scores to pkc@cs.fit.edu
- Score (0-10) for each member: circle a score (or circle two adjacent scores for .25 or write down a real number between 0 and 10)

Joanna Zhang	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
Gianni Bubb	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
Aidan Nelappana	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10

Faculty Advisor Signature: _____ Date: _____